

Jay Fukumoto

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EXPERIENCE

Software Development Engineer, Amazon Luna

Mar 2024 – Sep 2024

Amazon.com

Irvine, CA

- Refactored the controller metrics pipeline to surface data on previously untracked third-party controllers, raising labeled coverage by 30% and informing the decision to expand official controller support.
- Debugged and resolved TypeScript compatibility issues on unsupported client platforms to maintain baseline gameplay functionality outside official configurations.
- Authored an approved design for a client update notification system that polls build infrastructure to detect version drift and alert users running bundles outside the supported version window.

Software Development Engineer Intern, Amazon Luna

Jun 2022 – Sep 2022

Amazon.com

Irvine, CA

- Designed and implemented a cross-platform volume adjustment feature shipped to browser and macOS clients in TypeScript and Swift.
- Tested the backend UI configuration service to confirm feature changes propagated correctly across browser and macOS clients using Postman.
- Authored a formal requirements document reviewed by engineering and product stakeholders to lock scope before development began.

PROJECTS

HSRVision | *Next.js, TypeScript, Python* | hsr-vision.app

2026

- Built a browser-based real-time gameplay tracker that captures live game footage, classifies active characters using a fine-tuned ML model, and maintains a per-character damage leaderboard.
- Designed an OCR-driven state machine for real-time event detection, implementing a look-back capture to resolve a race condition between the detection system and the screen share.
- Built a two-stage auto-labeling pipeline using CLIP cosine similarity and logistic regression calibration to generate verified training data from 1M+ raw gameplay frames for the image classifier deployed for browser inference.

HSREndgameDB | *React, PostgreSQL, Node.js* | hsrendgamedb.info

2024 – Present

- Built a full-stack web app enabling players to upload and browse gameplay videos for challenge content in Honkai: Star Rail.
- Designed a mobile-first UI to match the game's primarily mobile player base, maintaining usability across desktop browser contexts.
- Implemented a split-table PostgreSQL schema separating post metadata from character data, keeping home page queries light while supporting full post assembly on the detail view.

SteamLens | *TypeScript, React, Express.js, Python*

2024 – 2025

- Built a Chrome Extension for researching game compatibility and reviews across ProtonDB, IGDB, and Steam.
- Developed Express.js API endpoints aggregating game data from Steam, IGDB, and ProtonDB into a single search response for the Chrome Extension.
- Built a Python ETL script to fetch, normalize, and cache the full 100k+ entry Steam game library, including alternate name generation and search optimization that improved latency and match accuracy by up to 25%.

EDUCATION

University of California, Irvine

Sep 2019 – Jun 2023

Bachelor of Science, Computer Science, GPA: 3.5

Irvine, CA

- Specialization in Information: coursework in information retrieval and database management.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, HTML/CSS, Java, Swift

Frameworks: React.js, Next.js, Express.js, Angular, Tailwind

Tools: Git, Jira, Postman, Vercel, Sentry, OpenSearch

Testing: Jest, Pytest

Databases: PostgreSQL, MySQL, NoSQL